Hashimoto’s Thyroiditis

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Microbiome Signatures, Microbiome Medicine

Abstract. Learn how Hashimoto’s thyroiditis (HT) connects to gut dysbiosis and toxic metals like nickel, arsenic, and lead.

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Overview

Hashimoto’s Thyroiditis (HT) is an autoimmune disorder characterized by the progressive destruction of thyroid follicles due to chronic inflammation, often leading to hypothyroidism. It affects 10-12% of the global population, with a significantly higher prevalence among women. While its etiology involves genetic, environmental, and epigenetic factors, increasing evidence highlights the role of gut microbiota in its pathogenesis. [1][2][3][4]

Research Feed

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Integrative analysis of gut microbiome and host transcriptome reveal novel molecular signatures in Hashimoto's thyroiditis.
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Zhao F, Feng J, Li J, Zhao L, Liu Y, Chen H, Jin Y, Zhu B, Wei Y.

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Thyroid. 2018

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Ishaq HM, Mohammad IS, Guo H, Shahzad M, Hou YJ, Ma C, Naseem Z, Wu X, Shi P, Xu J.

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A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.

Relative Abundance (Class)

A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.

Relative Abundance (Family)

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Relative Abundance (Genus)

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Relative Abundance (Kingdom)

A microbiome signature is defined through a comprehensive approach that integrates differential abundance analysis across multiple datasets, followed by interpretation for biological relevance. This rigorous process ensures that the identified signature is both statistically robust and biologically meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.

Relative Abundance (Order)

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meaningful. Associations that do not meet statistical significance or lack mechanistic relevance are excluded from the final signature.
Relative Abundance (Phylum)
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Relative Abundance (Species)

▶ **Studies**

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1. Integrative analysis of gut microbiome and host transcriptome reveal novel molecular signatures in Hashimoto's thyroiditis.. Li, M., Chen, K., Chen, Y. et al. Integrative analysis of gut microbiome and host transcriptome reveal novel molecular signatures in Hashimoto's thyroiditis.. (J Transl Med, 2024.)
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